SSSP HMW2: Wave Digital Filter modeling

Report for WDF model of piezoelectric MEMS loudspeaker

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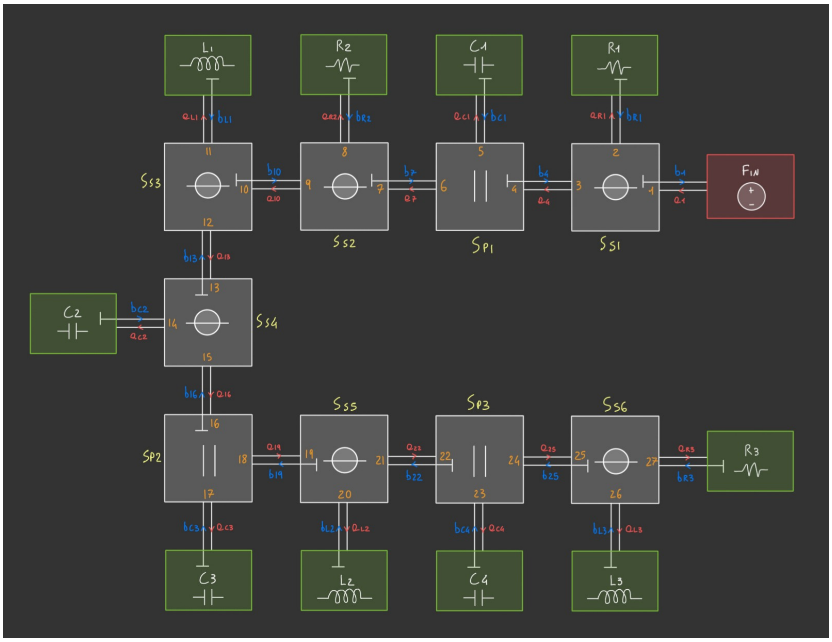

Figure 1: WDF scheme.

In our chosen WDF realization, each one-port (leaf) element is associated with a port impedance Z that must be set to ensure reflection-free behavior at a designated port of each adaptor. For leaf-port connections:

$$R_i \mapsto Z = R_i, \quad L_i \mapsto Z = \frac{2\,L_i}{T_s}, \quad C_i \mapsto Z = \frac{T_s}{2\,C_i}.$$

For an N-port series adaptor, choosing port 1 as the adapt port:

$$Z_1 \;=\; \sum_{n=2}^{N} Z_n\,,$$

For an N-port parallel adaptor, again choosing port 1 as the adapt port:

$$Z_1 \;=\; \Bigl(\sum_{n=2}^{N} Z_n^{-1}\Bigr)^{-1},$$

By applying these symbolic adaptation rules we obtain:

$Z_{27} = R_3,\quad Z_{26} = \frac{2\,L_3}{T_s},\quad Z_{25} = Z_{26} + Z_{27},\quad Z_{24} = Z_{25},\quad Z_{23} = \frac{T_s}{2\,C_4},$
$Z_{22} = \frac{Z_{23}\,Z_{24}}{Z_{23}+Z_{24}},\quad Z_{21} = Z_{22},\quad Z_{20} = \frac{2\,L_2}{T_s},\quad Z_{19} = Z_{20} + Z_{21},\quad Z_{18} = Z_{19},$
$Z_{17} = \frac{T_s}{2\,C_3},\quad Z_{16} = \frac{Z_{17}\,Z_{18}}{Z_{17}+Z_{18}},\quad Z_{15} = Z_{16},\quad Z_{14} = \frac{T_s}{2\,C_2},\quad Z_{13} = Z_{14} + Z_{15},$
$Z_{12} = Z_{13},\quad Z_{11} = \frac{2\,L_1}{T_s},\quad Z_{10} = Z_{11} + Z_{12},\quad Z_9 = Z_{10},\quad Z_8 = R_2,$
$Z_7 = Z_8 + Z_9,\quad Z_6 = Z_7,\quad Z_5 = \frac{T_s}{2\,C_1},\quad Z_4 = \frac{Z_5\,Z_6}{Z_5+Z_6},\quad Z_3 = Z_4,$
$Z_2 = R_1,\quad Z_1 = Z_2 + Z_3.$

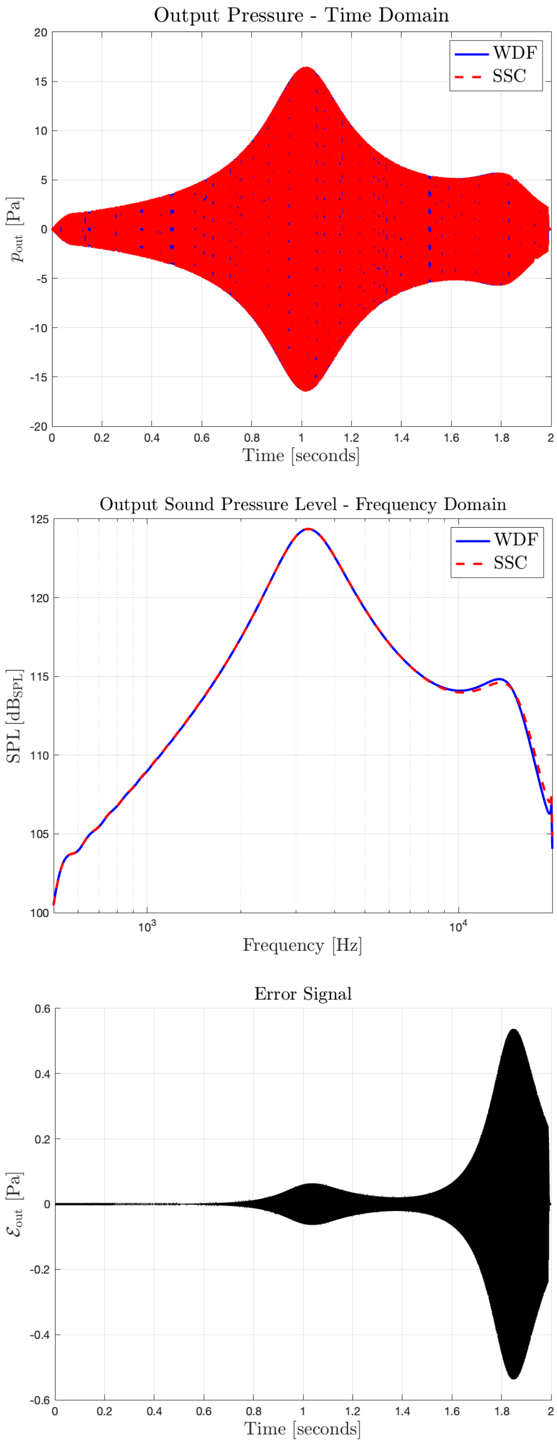

Figure 2: (a) Time-domain comparison with the ground truth. (b) Frequency-domain comparison with the ground truth. (c) Difference between ground truth and WDF output (**MSE** $=$ **0.0119**).